

Information Landscape Sculpting: Reconstructing Latent Epistemic Fields through Modern Associative Memory and Algorithmic Auditing

Revised draft (v2). Principal changes from v1: the central claim is recalibrated from “extract secret/censored research” to “reconstruct the adjacency structure and reading-front of a niche field,” the experimental protocol is upgraded to a heterogeneous multi-persona ensemble, a Limitations and Falsifiability section is promoted to the spine of the argument, and the ethics section is extended to address targeting assumptions and their failure modes.

Abstract of the Thesis

Commercial search and recommendation systems do not merely catalog discrete events. They project aggregate user behavior onto a low-dimensional shared embedding space in which cohesive cohorts (“birds of a feather”) settle into localized basins of attraction. This document argues that such systems are, mechanically, continuous associative memories, and that their basin geometry can be navigated by low-entropy, semantically adjacent agentic probing.

The claim recalibrated from v1 is important, because v1 overreached. The method **cannot** recover work that has left no behavioral trace in tracked commercial infrastructure — genuinely air-gapped or classified research is unrecoverable by construction. What it can plausibly do is *nowcast the frontier*: reconstruct the adjacency structure, terminology, and reading-front of a specialized field faster than lexical search surfaces it, by exploiting the aggregate lookalike dynamics the platform has already computed over real practitioners. This is a de-anonymization-of-the-frontier move, not an exfiltration-of-secrets move, and the distinction governs both what the method is good for and where it fails.

1. The Associative-Memory View of Retrieval

1.1 Two-Tower matching

Modern retrieval scales candidate selection by projecting users and items into a shared

vector space via a two-tower architecture. A user tower $f(x_u)$ maps user-side features (browsing history, real-time context, coarse location) into an embedding $u \in \mathbb{R}^d$; an item tower $\tilde{f}(\tilde{x}_i)$ maps item-side features (text, tags, ad creatives) into $v_i \in \mathbb{R}^d$. Interaction likelihood is the inner product or cosine similarity $\langle u, v_i \rangle$.

Offline, these representations are optimized with a contrastive objective such as InfoNCE, which pulls positive interaction pairs together and pushes sampled negatives apart:

$$L = -\log \left[\frac{\exp(\text{sim}(u, v^+)/\tau)}{\sum_j \exp(\text{sim}(u, v_j)/\tau)} \right]$$

with τ a temperature regulating distribution entropy. This forces the latent space to organize homophilously: users with similar histories cluster, and items consumed by a subpopulation map into the vicinity of that cohort’s user vector.

1.2 The Hopfield identity

Ramsauer et al. (2020) generalized the discrete Hopfield energy to continuous states with an exponential interaction term, yielding the update rule

$$\xi_{\text{new}} = X \cdot \text{softmax}(\beta \cdot X^T \cdot \xi)$$

where X is the matrix of stored patterns, ξ the state vector, and β the inverse temperature. Setting $Q = \xi^T$, $K = X^T$, $V = X^T$ and $\beta = 1/\sqrt{d_k}$ makes this **identical** to scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$$

The stored items of a recommender, or the factual parameters of a language model, are the fixed points of the system’s energy landscape.

1.3 The β phase transition (why sculpting is possible at all)

β sets the landscape topology. At high temperature (small β) the softmax is diffuse and the update converges to a single global fixed point averaging all patterns — generic, high-entropy content. As β increases, the landscape undergoes a phase transition, fracturing from one global basin into sharp local minima sitting on individual stored patterns (1-nearest-neighbor retrieval).

Component	Two-Tower Recommender	Transformer Attention	Continuous Hopfield
Query	Real-time user embedding u	Query Q	State vector ξ

Storage	Item matrix V	Key/Value K, V	Pattern matrix X
Scaling	Inverse temperature $1/\tau$	$1/\sqrt{d_k}$	Inverse temperature β
$\beta \rightarrow 0$	Uniform recommendation (max discovery)	Uniform attention	Single global attractor
$\beta \rightarrow \infty$	Nearest-neighbor retrieval (siloining)	Hard-max lookup	Sharp pattern retrieval

The practical obstacle is **popularity bias**: in graph collaborative filtering, highly connected nodes dominate message-passing aggregation, flattening the landscape and shallowing the basins of niche items. Any probing method must fight this flattening to reach an isolated minimum — a point that returns, with force, in the Limitations section.

2. Why Concept Deletion Is Incomplete (and the Limits of That Observation)

When developers sanitize a model or index — token banning, lexical filtering, closed-form unlearning — the underlying semantic relationships are rarely fully erased. Closed-form frameworks (e.g., CURE, which uses SVD over token embeddings to isolate a concept’s discriminative subspace and projects cross-attention weights orthogonal to it; representation-alignment editing such as RMU) suppress surface queries but do not destroy the distributed attractor. Representation-engineering work shows complex concepts are encoded as distributed, low-rank trajectories across residual streams; linear probes separate the classes cleanly at every layer, yet ablating an isolated direction often fails to change behavioral output. Once a forward pass initializes inside an attractor, it tends to commit to the semantic direction (“linguistic inertia”), and the concept survives removal of any localized subset of its encoding.

Honest scope note. This is an *interpretability* finding about the incompleteness of erasure. The inference from “erasure is representationally incomplete inside a model you can instrument” to “therefore a black-box commercial recommender’s public output can be steered to reconstruct a filtered concept” is a separate, much stronger claim, and it is not established here. v1 treated the two as equivalent; they are not. Section 5’s protocol should be read as a *test* of whether the recommender-level version holds, not as a demonstration that it does.

3. Physical Analog: Volumetric Photonic Computing (ImpCarv)

Implosion carving (ImpCarv), developed by Yang, Boyden, and So at MIT, is a photofabrication technique that specifies the refractive index at every coordinate of a material with nanoscale precision. A swollen polyacrylate hydrogel is immersed in a photosensitizing dye; a focused multi-photon laser scans internal coordinates, exciting the sensitizer to generate reactive oxygen species that cleave polymer bonds and create 3D vacancies. Ionic dehydration and supercritical drying then shrink the gel isotropically $\sim 10\times$ per dimension ($\approx 2,000\times$ volume), compressing 800 nm patterns to sub-100 nm features. Because these are smaller than the wavelength of visible light, the vacancies carry a distinct refractive index and diffract light in controlled, nonlinear ways.

The MIT group optimized the 3D vacancy pattern with deep learning to build an all-optical digit classifier: projected light propagates through the carved phase layers, performs a passive physical forward pass, and focuses onto the coordinate encoding the classification.

The analogy is structural and honest as such: propagation of light down a sculpted refractive-index gradient toward a stable focus is *isomorphic in form* to a query descending a sculpted behavioral landscape toward a stable basin. It illustrates that “landscape sculpting” is a real physical phenomenon; it is **not** evidence that the recommender-extraction claim works. The device proves landscapes have sharp basins, nothing more.

4. Experimental Protocol: Ensemble Agentic Probing

v1 specified a single-lane recursive loop. This is the upgraded design.

4.1 The core move

Do not replay any individual’s private trace. Instead, instantiate a **population of N synthetic personas**, each anchored to the *public* footprint of a distinct real practitioner in the target field, and each seeded with semantically adjacent, deliberately non-explicit terminology. Let the platform’s own lookalike model perform the clustering. Each persona is driven toward the individual attractor near its anchor; the *ensemble* is the instrument.

Why an ensemble rather than one profile:

- **Ensemble denoising.** Any single practitioner’s footprint carries idiosyncratic noise (unrelated interests, personal browsing). The *intersection* across N personas suppresses per-individual noise and leaves what is common to the cohort.
- **Graph recovery, not centroid recovery.** Sampling many points across the cohort recovers the field’s internal *structure* — sub-clusters, bridges, dense regions — rather

than a single averaged direction. The map is the valuable output, not a point on it.

- **Detection surface.** Counterintuitively, heterogeneity *helps*. Homogeneous parallelization (N clones of one persona) reads as a botnet; N behaviorally distinct personas are harder to fingerprint as coordinated. The residual tell is shared infrastructure (VM provider, correlated timing, a common extraction backend) — an OPSEC problem, not a design flaw, addressed in 4.4.

4.2 Per-persona initialization

Each persona runs in an isolated browser container (e.g., Playwright), maintaining identity through cookie state rather than verified accounts, so interactions leak to platform tracking from a clean slate. Each is driven by an LLM simulator with a specific, low-entropy technical persona and a seeded library of adjacent terms that avoid the explicit target phrasing — e.g., for a nanophotonics anchor:

- *Polymer chemistry:* acrylate hydrogel swelling, multi-photon photopatterning, reactive-oxygen-species kinetics.
- *Thermodynamics:* supercritical CO₂ drying, isotropic volume reduction, ionic dehydration.
- *Wave optics:* phase-contrast wavefront shaping, sub-diffraction index modulation, all-optical classification.

4.3 Steering and readout

Each persona executes targeted queries combining adjacent domains, then clicks and dwells **exclusively** on low-ranking, specific technical results, ignoring high-degree popular nodes to bypass popularity-bias aggregation. As the two-tower embedding is driven into a low-density region, the lookalike model groups the persona with the real cohort. The agent then harvests the shifted feeds, suggested items, and ad-network bids; a secondary multimodal model performs deductive coding to extract surfaced terms, identifiers, and citations; these re-seed the library and the cycle repeats. Across the ensemble, the pipeline reconstructs the cohort graph rather than a single trajectory.

4.4 OPSEC residual

Because heterogeneity defeats behavioral fingerprinting but not infrastructural correlation, a serious deployment must decorrelate provider, timing, and extraction path across personas. This is the dominant remaining detection vector.

Phase	Ensemble action	Platform response

Init	N containers; each persona seeded to a distinct anchor with adjacent, low-entropy terms	N clean guest IDs at latent-space origin
Steering	Targeted queries; dwell on niche nodes; ignore popular ones	Embeddings driven toward low-density cohort regions
Cohort match	Traverse redirected suggestions per persona	Lookalike grouping with the real cohort; local basin sharpens
Reconstruction	Scrape feeds; MLLM deductive coding; intersect across personas	Item delivery re-aligns to cohort behavior
Re-seed	Fold extracted terms back per persona; repeat	Increasingly specific adjacent suggestions

5. Limitations and Falsifiability (*the load-bearing section*)

This is where v1 was weakest and where an honest version must spend its credibility. The framework is elegant; whether it *works* rides almost entirely on assumptions that are currently untested.

5.1 The crux: recommender output signal-to-noise. Everything downstream assumes the platform's *output* — ad creatives, suggested items, feed reordering — carries enough signal to reconstruct "specific technical terms, patent IDs, institutional citations." That output is low-bandwidth, coarsely quantized, and (per §1.3) heavily popularity-biased. There is no established result that it encodes recoverable frontier specifics. This is the single load-bearing assumption. **Falsifiable test:** against a labeled ground-truth frontier (e.g., a field's arXiv reading-front over a fixed window), measure the mutual information between the injected persona and the recovered term set, and compare recovery precision/recall to a lexical-search baseline. If the ensemble does not beat lexical search on out-of-index terms, the thesis fails.

5.2 Trailing edge vs. leading edge. A target's *public* footprint anchors you to where they *were*. The frontier you want is what they are reading *now* — which by definition has not surfaced in their outputs; that is what makes it the frontier. The ensemble buys high-fidelity entry and clean readout; it does nothing for the *projection* step, which requires the platform to have ingested the cohort's *current* behavior and to forward-project it onto lookalikes. That is a stronger assumption than mere clustering, and it is the second place the method can silently fail. The design's implicit bet is honest and worth stating plainly: *I cannot know what they are reading now, but I can look enough like them that the algorithm shows me*

what it shows people like them now. Whether that pays is an empirical property of the platform's lookalike resolution.

5.3 Popularity bias fights back. The GNN aggregation the method must defeat actively re-centers profiles toward dense basins. Steering into a deep isolated minimum and *remaining* there against that pressure may be unstable; the basin you reach may decay back toward the cohort mean between cycles.

5.4 Sim2Real / simulator bias. LLM-driven personas carry linguistic and demographic biases. The behavioral distribution you inject may not match the real cohort's, landing you in an adjacent-but-wrong basin whose readout looks plausible but is off-target — a failure mode that is hard to detect from inside the loop.

5.5 Detection and terms of service. Automated sock puppets violate platform terms and risk flagging and banning (documented for SOAP-style active agents). Heterogeneity (§4.1) mitigates but does not eliminate this, and §4.4's infrastructural correlation remains.

5.6 What would convince a skeptic. A single end-to-end demonstration on a *benign, verifiable* target field — where the recovered frontier can be checked against a later-published ground truth — would move this from "reads well" to "holds." Absent that, the math (§1) and the physical analog (§3) establish only that landscapes have sharp basins, not that commercial recommender output has the resolution to reconstruct a research frontier. That inferential gap is the whole ballgame.

6. Ethical, Legal, and Targeting Considerations

6.1 The legitimate–abuse axis. The same protocol spans clearly legitimate uses (auditing recommenders for filter bubbles, radicalization pathways, or discrimination under the EU DSA and AI Act; competitive technical intelligence over genuinely public signals) and clear abuse (targeted surveillance or de-anonymization of identifiable individuals; circumventing bot-detection and terms of service; evading legitimate content controls). The distinguishing variables are consent, target specificity, and whether one is recovering *aggregate structure* or surveilling *identifiable people*. Recovering a field's terminology graph is categorically different from reconstructing what a named person is doing.

6.2 Targeting is an assumption the method does not license. The protocol is *origin-agnostic*: it sculpts whichever cohort the platform has already mapped. Framing a target cohort as a particular national or institutional apparatus is a targeting choice layered on top, and it should be scrutinized separately from the method. In particular, using a demographic proxy (inferred nationality or ethnicity) as a stand-in for either "the research frontier" or "a security category" conflates three distinct and successively smaller populations — surname

≧ national origin ≧ jurisdiction-bound national — and inherits a well-documented false-positive problem when origin is used as a security signal. Even granting a high $P(\text{demographic} \mid \text{adversary})$, the base rate makes $P(\text{adversary} \mid \text{demographic})$ small; the likelihood ratio runs the wrong way for individual-level inference, and operationalizing the reverse produces mostly false positives. The talent data can support “aim the probe at the AI-research cohort”; it does not support “and that cohort is a hostile apparatus.” Keeping the method at the level of *aggregate field reconstruction* avoids importing this error.

6.3 Privacy and the creep threshold. Even for aggregate targets, the technique instruments other people’s un-volunteered behavior. The tension between precision personalization and consumer privacy — the point at which accurate inference becomes a felt violation — applies to the auditor as much as to the platform being audited.

Appendix: Concept Map (updated)

graph TD

Root[Information Landscape Sculpting]:::root

Root --> RetrievalTopo[Retrieval Topology]:::mathematical

Root --> AssociativeMemory[Associative Memory]:::cognitive

Root --> Unlearning[Incompleteness of Erasure]:::cognitive

Root --> Photonics[Physical Analog: ImpCarv]:::physical

Root --> Auditing[Ensemble Probing]:::auditing

Root --> Limits[Limitations & Falsifiability]:::limits

RetrievalTopo --> TwoTower[Two-Tower u,v Embeddings]:::mathematical

RetrievalTopo --> GraphCF[Graph CF & Popularity Bias]:::mathematical

TwoTower --> Contrastive[InfoNCE Optimization]:::mathematical

AssociativeMemory --> ContinuousHopfield[Continuous Hopfield Networks]:::cogn

ContinuousHopfield --> AttentionIdentity[Identity to Attention]:::mathematica

ContinuousHopfield --> TemperaturePhase[Inverse-Temperature Phase Transition]

Unlearning --> Distributed[Distributed Residual-Stream Attractors]:::cognitiv

Unlearning --> ScopeNote[Scope Note: Interpretability != Black-Box Extraction]

Photonics --> ImpCarv[MIT Implosion Carving]:::physical

Photonics --> AnalogyNote[Analogy is Structural, Not Evidential]:::limits

Auditing --> Ensemble[Heterogeneous N-Persona Ensemble]:::auditing

Ensemble --> Denoise[Intersection Denoising]:::auditing

Ensemble --> GraphRecover[Cohort Graph Recovery]:::auditing

Ensemble --> OPSEC[Infrastructural Decorrelation]:::auditing

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Limits --> SNR[Crux: Recommender Output SNR]::limits
Limits --> Edge[Trailing vs Leading Edge]::limits
Limits --> Sim2Real[Sim2Real & Simulator Bias]::limits
Limits --> Targeting[Targeting Assumptions & Base Rates]::limits

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Conclusion

Predictive retrieval, associative-memory topology, and nanoscale optics converge on one structural fact: information settles into stable basins along paths of least resistance, and those basins can be navigated. The honest version of the thesis stops there and asks the empirical question rather than assuming its answer. Low-entropy ensemble probing *may* let an auditor reconstruct the adjacency structure and reading-front of a specialized field by borrowing the lookalike dynamics a platform has already computed over real practitioners. Whether it beats a lexical baseline on out-of-index terms — and whether the platform forward-projects a cohort's *current* behavior at all — are measurable questions, and they are the ones that decide whether any of this is real. The framework's value is that it makes those questions precise; its integrity depends on not pretending they are already settled.